



Weather and Climate



Learning Outcomes

The learner:

- defines weather and climate and explains the differences.
- outlines the factors that affect weather and climate.
- demonstrates sensitivity towards climate change.



Let's Take-off

Observation

Which of the following words describe this picture?

- | | | |
|---------|----------|---------|
| • hot | • sunny | • windy |
| • rainy | • cloudy | • cold |

It is a day.



C-1.3

DIFFERENCE BETWEEN WEATHER AND CLIMATE

The changes of the atmosphere of a particular place during a short period of time is called '**weather**'. Some days are sunny, some are windy and some days are cold. These changes over a short period of time are the weather of that place at that time.



Kerala in south India has a wet climate. Its houses have sloping roofs.



Ladakh in north India has a cold climate. Its houses are built to let in lots of sunlight.

The regular pattern of weather that a place has over a long period of time is called '**climate**'. If a place is usually hot with plenty of rain over several years, we say that the place has a hot and wet climate.

Weather can change over a short period of time, but climate remains constant over a several years. The clothes we usually wear and the food we eat depend on the climate.

Any change in any one of these factors, can cause the weather to change.



Discuss

Communication



How do changes in the weather affect you, your friends and your family? **C-1.3**



Think and Answer

Critical Thinking



Why do houses in Kerala have sloping roofs? **C-6.1**

FACTORS THAT AFFECT WEATHER

Temperature

Just as your body temperature can sometimes change, similarly the temperature of the atmosphere can be either hot or cold. This depends on how much heat it absorbs from the sun. Apart from heat absorption, pollution of different kinds can also make the atmosphere hot.

Clouds

Clouds are masses of water vapour that are formed when water evaporates from the earth's water bodies. Cloudy skies in the daytime make the day less hot, as all the heat from the sun does not reach the earth. But at night, clouds in the sky make the night warm as the trapped heat cannot escape through the clouds.

Wind

The movement of air in the atmosphere is called **wind**. How the wind affects us depends on the season. For example, in winters a strong wind makes us feel cold.

Humidity

When there is a lot of water vapour in the air, we say that the atmosphere is humid. Areas that are closer to water bodies feel warm and sticky when the air is **humid**.

Rain

When the clouds become very heavy with moisture, the water vapour condenses and falls as rain. Rain also affects weather. A lot of rain in a day makes the weather very wet and rainy.



HOTS

Critical Thinking



City A has been hot and sticky for a few days. What changes in weather can be expected due to this? **C-6.1**



Knowledge of India

Hindustani classical music has a raga especially devoted to the rains. It is called *Megh Malhar* (*megh* means cloud and *malhar* means downpour). It is believed that when it is played or sung, it has the power to bring showers of rain!



Did You Know?

Animals sense changes in weather better than humans. Frogs and insects become noisier before rain. Long before scientific ways to forecast the weather, early humans looked at natural signs for changes in the weather.

Any change in any one of these factors, can cause the weather to change.

FACTORS THAT AFFECT CLIMATE

Just as these factors make the weather change on a day to day basis, there are factors that affect the climate of a place:

Distance from the Equator

The equator at 0° gets direct rays of the sun. Places near the equator are the hottest. Andaman and Nicobar Islands are near the equator and they have a hot and humid climate. The further away from the equator a place is, the cooler it is. That is because of the sun's rays which are slanting and hence less heat is received.

Here is a temperature table for December at Jakarta and New York.

Which place has a colder climate? Why?

City	Latitude	Maximum	Minimum
Jakarta	6.17° S	32° C	24° C
New York	40.71° N	8° C	-1° C

Distance from the Sea

Land that is closer to the sea is influenced by it. Water heats and cools more slowly than land. Hence the land by the sea remains warmer in summer and colder in winter, than the land that is far away from the sea. Places far away from the sea are colder in winter and hotter in summer. That is why summer and winter in Delhi are much hotter and colder than in Mumbai, which is by the sea.

Altitude or Height of Land

The **altitude** of a place—its height above sea level—also affects its climate. Highlands like mountains and plateaus are much cooler than places where the land is at sea level. High places will have a much cooler climate



Discuss

Communication

21st
CENTURY
SKILLS

Why do many people go to hill towns during summer? Talk about any such places that you have been on holidays to.



Mt Kilimanjaro

even if they are close to the equator. Mt Kilimanjaro in Africa is 5,895 metres high and has snow at the top even though it is almost at 0° equator.

Direction of Winds

The direction of the regular winds that blow at a particular time of the year also affects the climate of a place. The **monsoon winds** that blow into the Indian subcontinent after blowing over the southern water bodies bring rain to most of India starting in June. The winds, which blow after crossing the Bay of Bengal in December and January, bring rain to the south-eastern coastal region of India.

Weather and climate are a result of geography, the earth's atmosphere and the location of a place. In the next chapter you will learn about the division of the earth into climatic belts.



A Better World

Sustainable Development Goals (SDGs)

Goal 13: Climate Action

Pollution caused by human beings is having a huge negative effect on climate. Human actions like pollution of air, water and land, and cutting of trees for commercial use are causing damage to our planet and unexpected changes are taking place in the climate of a place. What actions can we take in our daily life to prevent climate change?

C-2.3



Values and Life Skills

1. Our planet is our only home and is precious to us. We need to live in a responsible manner and avoid polluting it so that its environment and climate remain unspoilt. How can we do this?
2. What factors would you take into account while packing for a holiday to another city or country?

REMEMBER

Weather and Climate

Weather can change in a day or from day to day. This happens because of changes in the atmosphere.

Climate is the usual pattern of weather that a place has over a long time.

Several factors affect the weather and a climate of a place.

Weather and climate are all caused by the geography of a place, the earth's atmosphere and the location of a place..



NEW WORDS

Humid when there is a lot of water vapour in the air

Altitude the height of a place above sea level



EXERCISES

Remembering and Understanding

A. Choose the correct answer.

1. A place close to the equator is **hotter/cooler/wetter** than a place far away from the equator.
2. Places at a high altitude are **hotter/colder/wetter** than places at sea level.
3. In summer, places far away from the sea are **warmer/windier/wetter** than places that are beside the sea.
4. In winter, places beside the sea are **warmer/wetter/cooler** than places far away from the sea.
5. Weather can be **fixed/changing/unstable** every day.
6. Read the statements below and choose the correct option.

Statement (A): Multiple factors affect the climate of a place.

Statement (A): Climate is changing every year.

- (a) Both A and B are true, and B is the correct explanation of A.
- (b) Both A and B are true, but B is not the correct explanation of A.
- (c) A is true but B is false.
- (d) A is false but B is true.

B. Fill in the blanks.

1. Weather can change over a period of time.
2. is the usual weather a place has over a very long period of time.
3. Our lives are affected by both and
4. Places near the are the hottest.

C. Match the following.

A	B
1. Latitude	(a) Nice
2. Altitude	(b) A lot of water vapour in the air
3. Pleasant	(c) Parallel to the equator
4. Humid	(d) Height of a place above sea level

D. Write true or false. Correct the false statements in your notebook.

1. The weather can change from day to day.
2. Climate can change from day to day.
3. People build houses with sloping roofs in Ladakh.
4. The direction of winds at a particular time of the year affects climate.
5. Mumbai is hot and sticky because it is far away from the sea.

E. Answer the following questions.

1. Mention one important difference between weather and climate.
2. Give one example of how the climate affects the way people live.
3. Explain why the latitude of a place affects how hot or cold a place is.
4. Why does the sea affect the climate of a place?
5. How does temperature affect the weather of a place?



DO AND LEARN



A. Picture Study

Choose the picture you wish to talk about. Then, answer the questions below.

1. What kind of climate could these places have?
2. What kind of clothes would people usually wear?
3. What would some of the advantages and disadvantages be of living in such places?



B. On Your Own

C-2.3 Communication

- TP** 1. Keep a weather diary for a week. Make a table like this in your notebook and enter the day's weather every morning and evening. Some of these words may help you. You could use your own too.

Applying

Hot, warm, cold, cool, windy, wet, breezy, cloudy, humid

	Day 1	Day 2	Day 3	Day 4	Day 5	Day	Day 7
Morning							
Afternoon							
Evening							

2. Listen to raga *Megh Malhar*. Find out about other ragas that are associated with the weather or other forms of nature.

Research-based

C. In a Group

- TP** Work in small groups. Choose a bird, animal or insect. Make a PowerPoint presentation or poster explaining how they respond to changes in the weather or climate. Present the information to the other groups in class.

Collaboration

SI

Computer Science

- D. Pair and Share:** Along with your partner, discuss how altitude affects the climate of a place. Share the names of places that you know about or have been to.

C-6.1

Communication



Weblink For more information, go to:

- **Weather vs climate:** https://www.youtube.com/watch?v=_fANLICuYB8 (Accessed on 31 July 2025)



Self-review

What I Learnt

Colour the smileys to show how well you have understood the concepts.

Completely

Mostly

Need help

- I can define weather and climate and explain the differences.



- I can explain the factors that affect weather and climate.



- I can demonstrate sensitivity towards climate change.

